ALICE L WALTON SCHOOL OF MEDICINE

Antiplatelet Drugs

Last updated April 8, 2025 · 12 min read · ScholarRx

Adapted by: Trager Hintze, Diptiman Bose

Editors: Jordan Finn, Russell Johnson

Adapted from: Antiplatelet Drugs

Listen to this Brick
0:00 / 22:01 1x

Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Justify the use of pharmacological agents in treating **diseases associated with atherosclerosis** based on their mechanism of action and pharmacokinetic profile.
2. Recognize adverse drug reactions for medications used for **diseases associated with atherosclerosis.**
3. Apply an evi [illegible] oach to treat **diseases associated with atherosclerosis.**

CASE CONNECTION

EG is a 68-year-old male who was transported via ambulance to the emergency department for substernal chest pain. "I was sitting at home watching TV, and the pain hit me. It's now also in my left arm. I think I'm having a heart attack." You review the paramedic's record and order a stat ECG; it reveals an ST elevation in the inferior leads. EG is rushed to the cardiac catheterization lab for percutaneous coronary intervention (PCI), and two stents are placed in occluded coronary arteries.

Which medications will you prescribe to reduce EG's risk of myocardial infarction (MI) and death? Which medications will you prescribe to prevent stent blockage? Which medications should EG receive once the diagnosis of ST-elevation MI (STEMI) has been made, before transferring him to the cardiac catheterization lab? Consider your answers as you read, and we'll revisit EG at the end of the brick.

GO TO CONCLUSION ↓

Antiplatelet drugs are a class of pharmaceutical agents that inhibit the formation of pathologic blood clots (thrombi) and are particularly useful for treating arterial thrombi. Remember that arterial thrombi can lead to myocardial infarctions (MIs) and strokes, both of which are leading causes of mortality. Atherosclerosis causes MIs and strokes by clogging the arteries with plaques. Atheroscler [illegible] atelet plug formation, activation of the coagulation cascade, blockage of the involved vessel, and ischemia or infarction of downstream tissues, such as the myocardium or brain.

These antiplatelet agents differ according to where they act in the clot formation pathway. First, we'll review briefly how platelets form a plug. Then, we'll explore how individual antiplatelet drugs block different points in this platelet pathway to inhibit platelet function.

How Do Antiplatelet Drugs Work? (LO#1)

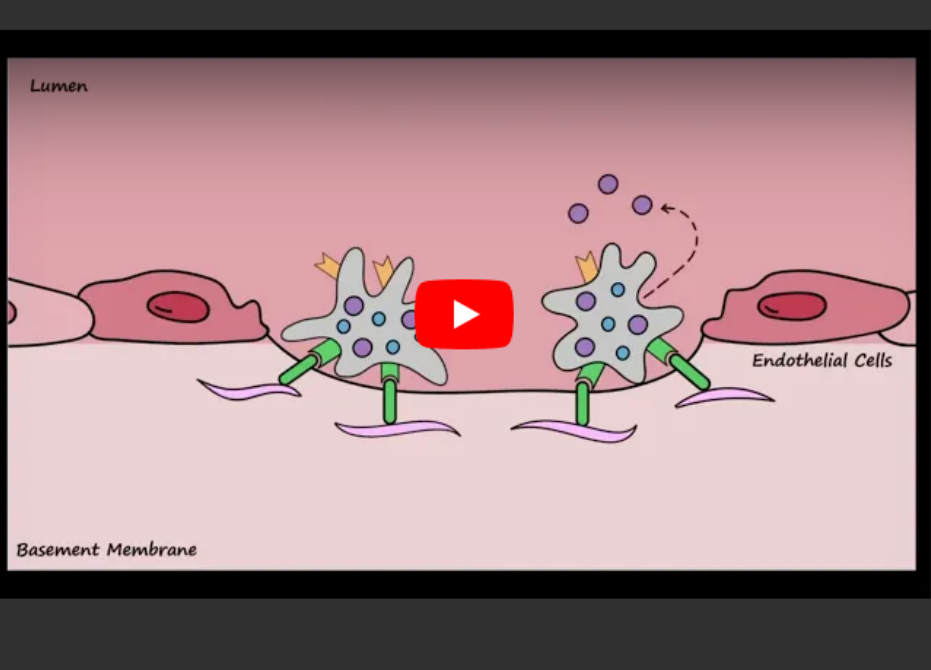
The common antiplatelet agents all inhibit clot formation by **blocking platelet aggregation**, but they do so in different ways. There are four different medication classes we will discuss:

- Cyclooxygenase (COX) inhibitor acetylsalicylic acid (i.e., ASA/ aspirin)
- $P2Y_{12}$ inhibitors
  - Clopidogrel, prasugrel, ticagrelor, and cangrelor
- Glycoprotein (Gp)IIb/IIIa inhibitors
  - Abciximab, eptifibatide, and tirofiban
- Phosphodiesterase-3 (PDE3) inhibitors
  - Dipyridamole and cilostazol

Before we look at the specific drugs, let's review the process of platelet plug formation. The video below describes this process.

Here is how a platelet plug forms:

- After endothelial injury, glycoprotein Ib (GpIb) receptors on the platelet surface bind to th [illegible] sing von Willebrand factor (vWF) as th [illegible]. This step is called **adhesion.**
- Platelets then secrete thromboxane A2 (TXA_2) and ADP, which recruit more platelets to the injured area. TXA_2 stimulates the platelet TP receptor and ADP stimulates the platelet $P2Y_{12}$ receptor, leading to increased expression of glycoprotein IIb/IIIa (GpIIb/IIIa) receptors on the platelet surface. This step is called **activation**.
- Finally, GpIIb/IIIa receptors on the surface of adjacent platelets bind to each other using fibrinogen as their glue. This final step is called **aggregation.**



How do antiplatelet drugs inhibit the formation of clots?

How do each of the antiplatelet medications work? (LO#1)

Cyclooxygenase Inhibitors

The most common antiplatelet medication you will encounter is aspirin. Recall from Foundations of Pharmacology in the Nonopioid Analgesic - Aspirin brick that aspirin is classified as a non-selective COX inhibitor and as a salicylate (Figure 1). Aspirin also inhibits platelet aggregation by the inhibition of thromboxane- A2 (TxA2). For diseases associated with atherosclerosis, this is the mechanism we care most about.

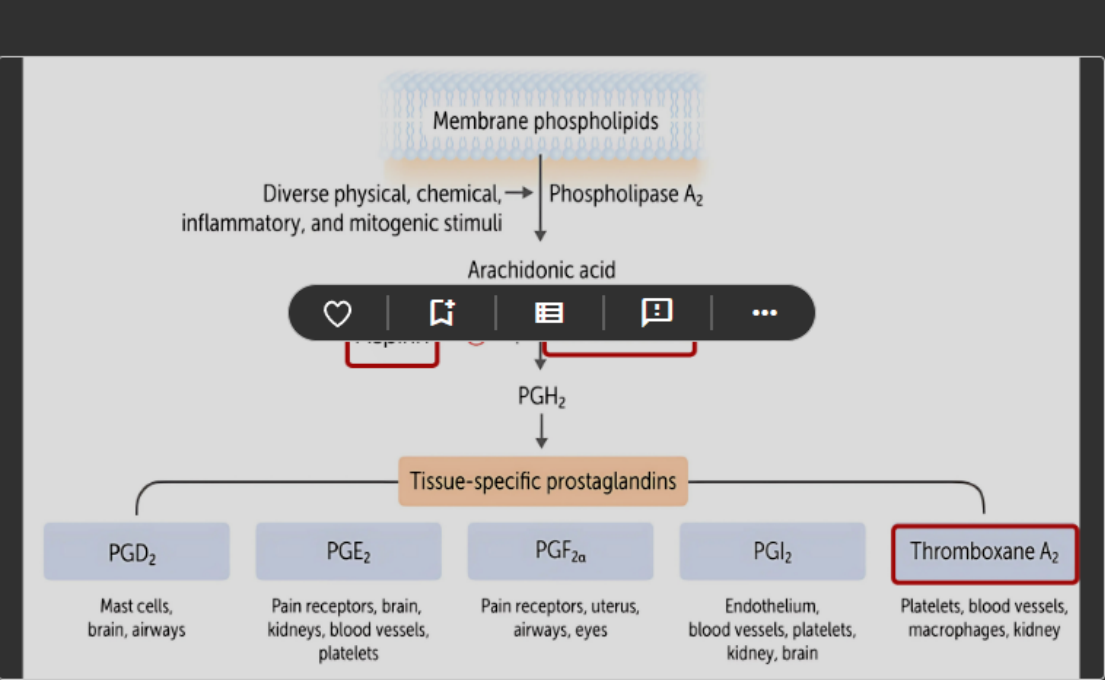

Membrane phospholipids
Diverse physical, chemical, inflammatory, and mitogenic stimuli
Phospholipase A2
Arachidonic acid
PGH2
Tissue-specific prostaglandins
PGD2
PGE2
PGF2α
PGI2
Thromboxane A2
Mast cells, brain, airways
Pain receptors, brain, kidneys, blood vessels, platelets
Pain receptors, uterus, airways, eyes
Endothelium, blood vessels, platelets, kidney, brain
Platelets, blood vessels, macrophages, kidney

Figure 1

Indications for aspirin (LO#3)

When should we use ASA as an antiplatelet agent? The principal indication is in the management of patients with **cardiovascular disease**, including myocardial infarction (MI), peripheral artery disease, and stroke. Additional indications are for patients with valvular disease or angina pectoris (stable or unstable), or who have had coronary artery bypass surgery. In all these cases, ASA is a crucial agent for **secondary prevention** of further ischemic events (e.g., repeat MI or stroke) in these patients. It's also part of the initial emergency department management for patients with chest pain who are suspected to be having an MI (acute coronary syndrome).

What is the r [illegible] vent platelet aggregation?

Adverse Effects (LO#2)

Common side effects of ASA therapy include dyspepsia, peptic ulcer, erosive gastritis, and *upper gastrointestinal bleeding*. Most often, bleeding due to aspirin can be managed by simple discontinuation and symptom management, however, in severe cases blood transfusions may be warranted.

At high doses, ASA has effects on the cochlea that can result in tinnitus. ASA overdoses can cause respiratory alkalosis (hyperventilation) resulting from stimulation of the central nervous system respiratory center, as well as lactic acidosis.

ASA can cause a pseudoallergic reaction, particularly in patients with a history of nasal polyps or asthma. This is because COX inhibition leads to a compensatory increase in LOX production, a metabolic intermediary along with COX and TXA_2 in the arachidonic acid metabolism pathway. Increased LOX leads to increased production of leukotrienes, important immune mediators that may induce asthma symptoms (wheezing) in susceptible patients.

CLINICAL CORRELATION

Aspirin for primary prevention of heart attack and stroke? Aspirin 81 mg, or commonly referred to as 'baby aspirin', is available over-the-counter in the United States. For many years there was a recommendation that patients at high risk for cardiovascular events take a 'baby aspirin' daily for prevention. In 2019, a large landmark trial (ASCEND) was published that showed higher rates of bleeding than rates of preventing heart attack and stroke in patients taking aspirin for primary prevention, i.e., the risk is greater than the benefit. So, the recommendation now is to only take a daily aspirin for secondary prevention, i.e., after a heart attack or stroke has occurred.

Contraindications (LO#2)

ASA is contraindicated in patients with a history of ASA-related anaphylactic reaction, *active or recent gastrointestinal bleeding or gastric ulcers*, recent intracranial bleeding, bleeding disorders, and severe liver or renal disease.

ASA should not be used in febrile children because of a concern for *Reye syndrome*, which is rare but potentially fatal. This can cause rapid fatty liver degeneration leading to hepatic encephalopathy and potentially irreversible brain damage. **Reye syndrome can occur when ASA is given to children during viral infections**.

What may happen if aspirin is given to a febrile child?

$P2Y_{12}$ Inhibitors (LO#1)

$P2Y_{12}$ inhibitors are also agents that inhibit platelet aggregation. These agents inhibit the $P2Y_{12}$ receptor for ADP, inhibiting platelet aggregation induced by platelet-secreted ADP. They block the expression of Gp IIb/IIIa on the platelet surface (see Figure 2). They include the irreversible inhibitors clopidogrel and prasugrel as well as the reversible inhibitors ticagrelor and cangrelor.

Knowing which of the P2Y12 inhibitors are irreversible vs irreversible is important because it plays a role in determining how long the platelets inhibited after the medication is discontinued. For instance, the irreversible inhibition of clopidogrel and prasugrel mean that they will have inhibitory effects on platelets up to 7 and 9 days, respectively. On the other hand, the reversible ticagrelor and cangrelor will only have effects for 3 days or immediately after discontinuation, respectively.

What is the mechanism of action of P2Y12 inhibitors?

Indications (LO#3)

Most often the P2Y12 inhibitors are used in combination with ASA for first-line **secondary stroke prevention** in patients with a past stroke or MI. Dual antiplatelet (DAP) therapy with ASA and a $P2Y_{12}$ blocker is recommended for patients for 1 year after percutaneous coronary intervention (PCI; coronary angioplasty with stenting) to prevent coronary stent occlusion. Also, like ASA, these drugs also are used in initial management of acute coronary syndrome.

P2Y12 inhibitors can also be used as alternative antiplatelet agents for secondary prevention in ASA-allergic patients.

Adverse Effects (LO#2)

Side effects are less common than with ASA. As many as 6% of patients taking clopidogrel may have a hypersensitivity reaction, marked by a skin rash appearing 1 week after starting treatment. However, given their effect on platelet aggregation, there is always an associated risk of bleeding for patients on these medications.

Contraindications (LO#2)

The drug is contraindicated in patients with severe liver disease. This is because clopidogrel is a prodrug that requires conversion to an active metabolite by CYP2C19 in the liver, so its effectiveness is reduced in patients with severe liver disease. This is also a consideration for other medications that are inducers/ inhibitors of 2C19.

Why is clopidogrel contraindicated in patients with severe liver disease?

Gp IIb/IIIa Inhibitors (LO#1)

These agents prevent platelet aggregation by binding to the Gp IIb/IIIa receptors on activated platelets (see Figure 2). Abciximab (a monoclonal antibody) and the synthetic compounds eptifibatide and tirofiban are Gp IIb/IIIa inhibitors tha [illegible] three drugs are given intravenously.

Indications (LO#3)

The Gp IIb/IIIa inhibitors are not used often, and they are only used for a small and specific subset of patients. They can be used in the early management of ACS for patients who will undergo PCI in which the coronary artery thrombus is mechanically disrupted using a catheter inserted into the coronary arteries. Eptifibatide and tirofiban are also used in some high-risk patients with unstable angina.

What is the mechanism of action of abciximab?

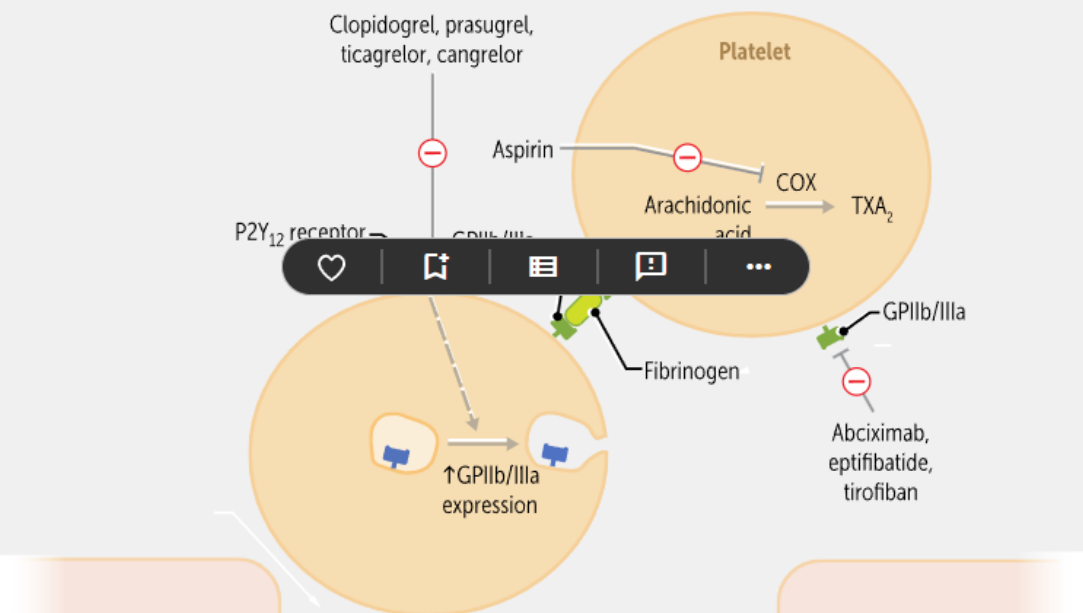
Adverse Reactions (LO#2)

The most common complication is bleeding, including retroperitoneal, pulmonary, and spon [illegible] arterial access site for the cardiac catheterization is an especially common site for bleeding. Risk factors for bleeding include older age, a history of bleeding disorders, and concomitant use of drugs that increase the risk of bleeding (anticoagulants, NSAIDs, and/or P2Y12 inhibitors). Patients <70 kg may be at greater risk for major and minor bleeding. Minimizing invasive procedures, including arterial and venous punctures, IM injections, and the use of urinary catheters, nasotracheal intubation and nasogastric tubes is an important measure to prevent bleeding in these patients.

Thrombocytopenia is another concerning adverse effect associated with Gp IIb/IIIa use. Acute, profound thrombocytopenia (immune-mediated and nonimmune mediated) has occurred and may occur within 24 hours of initiation. Platelet counts should recover rapidly (within 1-5 days) after discontinuation. Use with extreme caution in patients with platelet counts $<100,000/mm^3$.

Contraindications (LO#2)

Contraindications include active internal bleeding, history of recent surgery or trauma, stroke within the past 30 days, hemorrhage stroke or cranial neoplasm (at any time), cirrhosis, or thrombocytopenia.



QUIZ

Figure 2

Phosphodiesterase-3 Inhibitors (LO#1)

These agents inhibit phosphodiesterase, thereby increasing the concentration of cyclic adenosine monophosphate (cAMP) in platelets, which inhibits platelet aggregation (see Figure 1). PDE-3 inhibitors are now the least commonly used anticoagulants, as they have been largely replaced by aspirin and the P2Y12 inhibitors. They include dipyridamole and cilostazol.

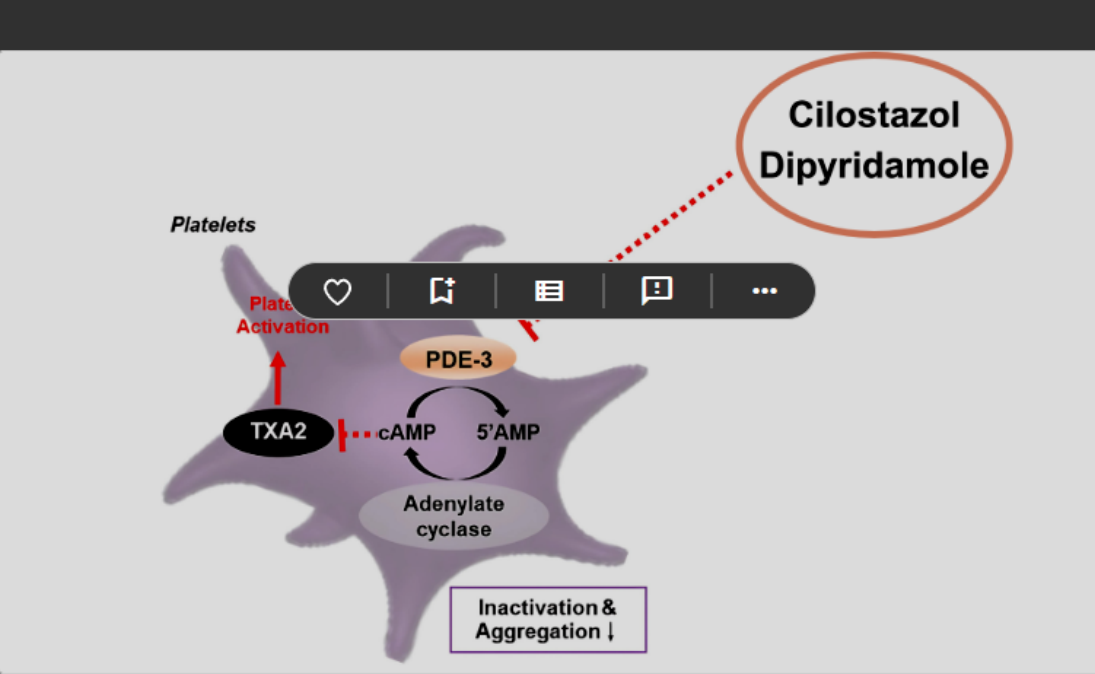


Figure 3

Indications (LO#3)

As mentioned above, these medications are not commonly used anymore. However, both of these medications can be used in patients with symptomatic peripheral artery disease with intermittent claudication. Dipyridamole may be used with warfarin to prevent clotting of artificial heart valves.

Adverse reactions (LO#2)

Remember the role of phosphodiesterases in the nitric oxide pathway? Both of these agents may produce adverse effects like flushing and headache because of their vasodilatory effects.

Contraindications (LO#2)

Cilostazol is contraindicated in patients with heart failure. Again, due to the effects of blocking PDE-3, it may cause profound hypotension in a patient with heart failure.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to EG, what medication do you prescribe to reduce EG's risk of death and MI? Which medications do you prescribe to prevent stent blockage?

You visit EG before he is discharged, 4 days after his stent placement. You tell him he will be given a discharge packet that includes prescription instructions. Included among the meds are ASA and an ADP receptor blocker. You explain to EG that both medications inhibit platelet aggregation, which if it occurs could reduce blood flow to his heart muscle and clog his stents. He agrees to take all medications exactly as prescribed and to follow up with the cardiologist in 2 weeks. You review with him the potential adverse effects to these medications, including unusual bleeding.

Summary

- Antiplatelet drugs inhibit the formation of pathologic blood clots by preventing platelet aggregation, crucial for treating arterial thrombi and preventing myocardial infarctions and strokes.
- Aspirin, a cyclooxygenase inhibitor, prevents platelet aggregation by inhibiting thromboxane A2 and is used for secondary prevention of cardiovascular diseases.
- P2Y12 inhibitors, such as clopidogrel and prasugrel, inhibit platelet aggregation and are used in combination with aspirin for secondary stroke prevention and post-PCI management.
- Gp IIb/IIIa inhib [illegible] telet aggregation by targeting specific receptors on activated platelets.
- Phosphodiesterase-3 inhibitors, like dipyridamole and cilostazol, increase cAMP in platelets to inhibit aggregation and are less commonly used, but can be indicated for peripheral artery disease.
- Common adverse effects of antiplatelet drugs include bleeding risks, and specific contraindications exist for each drug class, such as aspirin in children with viral infections due to Reye syndrome risk.

Review Questions

1. Which antiplatelet drug is a prodrug that requires conversion to an active metabolite by CYP2C19 in the liver?

A. ○ Aspirin
B. ● Clopidogrel
C. ○ Prasugrel
D. ○ Abciximab
E. ○ Cilostazol

Hide Explanation

The correct answer is clopidogrel (B). Clopidogrel is a prodrug that requires conversion to an active metabolite by CYP2C19 in the liver. Aspirin (A), prasugrel (C), abciximab (D), a [illegible] by CYP2C19 in the liver to become active.

2. Which antiplatelet drug is contraindicated in febrile children due to the risk of Reye syndrome?

A. ● Aspirin
B. ○ Clopidogrel
C. ○ Prasugrel
D. ○ Abciximab
E. ○ Cilostazol

Hide Explanation

The correct answer is aspirin (A). Aspirin is contraindicated in febrile children due to the risk of Reye syndrome, a rare but potentially fatal condition that can cause rapid fatty liver degeneration leading to hepatic encephalopathy and potentially irreversible brain damage. Clopidogrel (B), prasugrel (C), abciximab (D), and cilostazol (E) are not associated with Reye syndrome.

3. Which antiplatelet drug inhibits platelet aggregation by binding to the Gp IIb/IIIa receptors on activated platelets?

A. ○ Aspirin
B. ○ Clopidogrel
C. ● Abciximab
D. ○ Prasugrel
E. ○ Cilostazol

Hide Explanation

The correct answer is abciximab (C). Abciximab is a monoclonal antibody that inhibits platelet aggregation by binding to the Gp IIb/IIIa receptors on platelets. Aspirin (A) inhibits platelet aggregation by inhibiting thromboxane A2. Clopidogrel (B) and prasugrel (D) are P2Y12 inhibitors that inhibit platelet aggregation by inhibiting the P2Y12 receptor for ADP. Cilostazol (E) is a phosphodiesterase-3 inhibitor that inhibits platelet aggregation by increasing the concentration of cAMP in platelets.

How would you rate this Brick?

👍 👎

References

- https://academic.oup.com/esc/book/35489/chapter-abstract/312405367?redirectedFrom=fulltext
- https://www.tandfonline.com/doi/abs/10.1080/17512433.2022.2121702?journalCode=ierj20
- Kitchens CS, Kessler CM, Konkle BA, et al, eds. *Consultative Hemostasis and Thrombosis*. 4th ed. Elsevier; 2019.
- Michelson AD, ed. *Platelets*. 4th ed. Academic Press;
- Stevens CW. *Brenner and Stevens' Pharmacology*. 6th ed. Elsevier; 2023.
- Wecker L, Taylor DA, Theobald RJ, ed. *Brody's Human Pharmacology*. 6th ed. Elsevier; 2019.

Review Learning Objectives ✓

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Justify the use of pharmacological agents in treating diseases associated with atherosclerosis based on their mechanism of action and pharmacokinetic profile.~~
02 ☑ ~~Recognize adverse drug reactions for medications used for diseases associated with atherosclerosis.~~
03 ☑ ~~Apply an evidence-based pharmacotherapeutic approach to treat diseases associated with atherosclerosis.~~

abstract/312405367?redirectedFrom=fulltext
- https://www.tandfonline.com/doi/abs/10.1080/17512433.2022.21702?journalCode=ierj20
- Kitchens CS, Kessler CM, Konkle BA, et al, eds. *Consultative Hemostasis and Thrombosis*. 4th ed. Elsevier; 2019.
- Michelson AD, ed. *Platelets*. 4th ed. Academic Press;
- Stevens CW. *Brenner and Stevens' Pharmacology*. 6th ed. Elsevier; 2023.
- Wecker L, Taylor DA, Theobald RJ, ed. *Brody's Human Pharmacology*. 6th ed. Elsevier; 2019.

Review Learning Objectives ✓

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Justify the use of pharmacological agents in treating diseases associated with atherosclerosis based on their mechanism of action and pharmacokinetic profile.~~
02 ☑ ~~Recognize adverse drug reactions for medications used for diseases associated with atherosclerosis.~~
03 ☑ ~~Apply an evidence-based pharmacotherapeutic approach to treat diseases associated with atherosclerosis.~~

Objective Confidence: 3/3